Project 2: Turbojet Engine Design/Off-Design Performance Analysis

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The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

Nomenclature

T = temperature $[K]$
P = pressure $[Pa]$
ρ = density $\left[kg/m^3\right]$
T_t = total temperature $[K]$
P_t = total pressure $[Pa]$
τ = $\frac{T_{t,out}}{T_{t,in}}$
π = $\frac{P_{t,out}}{P_{t,in}}$
η = isentropic efficiency
u = velocity $[m/s]$
M = mach number
A = area $\left[m^2\right]$

Subscripts

0 = freestream
d = diffuser
c = compressor
b = burner
t = turbine
n = nozzle

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I. Introduction

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II. Completely define the on-design engine fluid dynamics and performance

A. Methodology

B. Results

III. Develop the complete performance envelope for this specific turbojet engine

A. Methodology

B. Results

1. Sea Level

2. 4000 m

3. 8000 m

4. 12000 m

Table 1 On-Design Operation

		Station						
		0	1	2	3	4	5	8
u	m/s	261.87	261.87	261.87	0	0	0	0
P	Pa	35651	35651	0	0	0	0	0
T	K	236.23	236.23	0	0	0	0	0
M		0.85	0.85	0.50	1.0			1.0
ρ	kg/m^3	0	0	0	0	0	0	0
A	m^2	0.00318	0.00318	0	0	0	0	0
P_t	Pa	0	0	0	0	0	0	0
T_t	K	0	0	0	0	0	0	0

IV. Conclusion

Appendix